Geometry Unit 2 Assessment Guide

General Information	This unit gives the student a general foundation in logic and proof. "Instruction utilizes different ways students can organize their reasoning by constructing various proofs when proving geometric statements. It is important to explain the terms statements and reasons, their roles in a geometric proof, and how they must correspond to each other. Regardless of the style, a geometric proof is a carefully written argument that begins with known facts, proceeds from there through a series of logical deductions, and ends with the statement you are trying to prove." Geometry B.E.S.T. Instructional Guide for Mathematics page 14. Unit 2 has students begin their journey with proofs by introducing them to the work of the father of Geometry, Euclid. By exploring a visual proof of vertical angles students have an manageable entry into following a series of logical deductions. Students build their knowledge of the benchmarks MA.912.LT.4.3 and MA.912.LT.4.10 through the incorporation of quotes and the UNO game. Gradually students use the knowledge of angles and line segments to complete proofs using a variety of formats. Since student understanding of proofs will grow, this assessment does not assess proofs at the same depth and cognitive demand as the student experience within lessons 7 and 8. Students are assessed on their knowledge of vertical angles using both relationships in a diagram and a proof. Being able to easily distinguish vertical angles becomes important prerequisite knowledge for unit 3 where students use vertical angles to make connections of angles formed when a transversal crosses parallel lines. In unit 2 students are introduced to the definition of congruence and the difference between congruent and equal. "Students should understand the difference between congruent and equal." Geometry B.E.S.T. Instructional Guide for Mathematics page 14. Questions 4 and 10b assess a student's understanding of this relationship. As understanding this subtle difference is utilized throughout Geometry, teachers should note if their students seem confused on the difference. Students should also have prior knowledge of this difference from Grade 8 or Grade 7 Accelerated Math.
Calculator Information	All questions are calculator neutral. Students may be given the use of a calculator and it will not have influence on their ability to show their knowledge of the content.
Total Recommended Points	The total recommended points in this guide is 100 points.

Question	1	Benchmark(s)	MA.912.LT.4.10
Inductive		Recommended Scoring (4 Total Points)	No partial credit is suggested.
Question	**2**	**Benchmark(s)**	MA.912.LT.4.10
Deductive		Recommended Scoring (4 Total Points)	No partial credit is suggested.
Question	**3**	**Benchmark(s)**	MA.912.LT.4.10
Deductive		Recommended Scoring (4 Total Points)	No partial credit is suggested.
Question	**4**	**Benchmark(s)**	MA.912.LT.4.10
Definition of Congruence		Recommended Scoring (4 Total Points)	No partial credit is suggested.
Question	**5a**	**Benchmark(s)**	MA.912.LT.4.10
Step **3**		Recommended Scoring (4 Total Points)	No partial credit is suggested.

Question	5b	Benchmark(s)	MA.912.LT.4.10
Yellow 7 or Red 3		Recommended Scoring (4 Total Points)	No partial credit is suggested.
Question	6a	Benchmark(s)	MA.912.LT.4.3
If I eat ice cream, then it is my birthday.		Recommended Scoring (4 Total Points)	No partial credit is suggested.
Question	6b	Benchmark(s)	MA.912.LT.4.3
Answers will vary. I eat ice cream on my brother's birthday.		Recommended Scoring (4 Total Points)	Partial credit is at the discretion of the teacher.
Question	7	Benchmark(s)	MA.912.LT.4.3
(see table below)		Recommended Scoring (8 Total Points)	Consider giving students 4 points for each correct if-then statement. Students may write the if-then statement slightly differently. Partial credit is at the discretion of the teacher.

Statement	If-Then
All Floridians have been to the beach.	If someone is a Floridian, then they have been to the beach.
All four angles at the point of intersection of two lines are right angles.	If two lines intersect, then all four angles at the point of intersection are right angles.

Question	8	Benchmark(s)	MA.912.LT.4.3
Hypothesis: Anthony has visited the Southernmost Point Buoy. **Conclusion**: Anthony has been to Key West.		**Recommended Scoring** **(8 Total Points)**	Consider giving students 4 points for a correct hypothesis and a correct conclusion. Students may write each statement slightly differently. Partial credit is at the discretion of the teacher.
Question	**9**	**Benchmark(s)**	MA.912.LT.4.3
Two angles are supplementary if and only if their sum is $180°$.		**Recommended Scoring** **(4 Total Points)**	Students may write the statement slightly differently. Partial credit is at the discretion of the teacher.
Question	**10a**	**Benchmark(s)**	MA.912.GR.1.1
☐ $\angle DRH \cong \angle KRC$ ☐ $\angle HRC \cong \angle KRD$ ☐ $\angle MKP \cong \angle BKN$ ☐ $\angle NKB \cong \angle RKP$ ☐ $\angle RDK \cong \angle RKD$ ☐ $\angle WDY \cong \angle RDK$		**Recommended Scoring** **(12 Total Points)**	Consider giving 4 points for each correct answer and deducting 4 points for each incorrect answer.

Question	10b	Benchmark(s)	MA.912.GR.1.1
(see proof table below)		Recommended Scoring (12 Total Points)	Consider giving students 4 points for each correct reason.

Statement	Reason
1. $m\angle MKB + m\angle BKN = 90°$	1. Given
2. $\angle MKB \cong \angle RKD$	2. Vertical Angles Theorem
3. $m\angle MKB = m\angle RKD$	3. Definition of congruence
4. $m\angle RKD + m\angle BKN = 90°$	4. Substitution

Question	10c	Benchmark(s)	MA.912.LT.4.3
(see statement table below)		Recommended Scoring (12 Total Points)	Consider giving students 4 points for a correct statement. Students may write each statement slightly differently. Partial credit is at the discretion of the teacher.

Relationship to Conditional	Statement
Converse	If $m\angle MKB + m\angle BKN = 90°$, then $m\angle MKN = 90°$.
Inverse	If $m\angle MKN \neq 90°$, then $m\angle MKB + m\angle BKN \neq 90°$.
Contrapositive	If $m\angle MKB + m\angle BKN \neq 90°$ then $m\angle MKN \neq 90°$.

Question	10d	Benchmark(s)	MA.912.LT.4.3
Both the statement and the converse are true.		**Recommended Scoring** **(4 Total Points)**	Consider giving students 4 points for a correct explanation. Students may write their explanation slightly different. Partial credit is at the discretion of the teacher.
Question	**11**	**Benchmark(s)**	MA.912.LT.4.10
Student answers will vary. Teacher will have to verify each student's answer.		**Recommended Scoring** **(8 Total Points)**	Consider giving students 2 points for each correct statement and 2 points for each correct reason.